Azure Capacity & Quota Management — Consolidated Requirements Baseline

> **Working document — coauthored baseline for the SaaS Design Services capacity & DR programme.** This is a living requirements document. It captures the agreed architectural direction while explicitly separating confirmed baseline requirements from items that remain configurable, need proof-of-concept (POC) validation, await a business decision, or depend on an external Azure capability.

Document Control

Field	Value
Title	Azure Capacity & Quota Management — Consolidated Requirements Baseline
Version	**4.0 — unified clean-slate rewrite (harmonises all v2.x carried-forward text with the v3.0 multi-domain customer model; unified formula reference; removes legacy single-scope ambiguity)**
Status	Working baseline — single self-consistent voice; no legacy amendment layers
Baseline date	14 October 2026 (supersedes v3.0 / 8 Oct 2026)
Owners	Vishnuvardhan Reddy (design/requirements), Roy Szabady (strategy/business alignment)
Contributors	Azure Platform Support (Anu — deployment pipeline), Jason (quota tooling/code), Melvin Stephen (CVP — quota-as-governor strategy)
Primary consumers	Exosphere, Stratosphere, AEP, product platform teams, operations, FinOps, security & governance
Collaboration space	Atlassian Confluence (space: **PI**, folder Azure Capacity Reservation) — chosen as the shared coauthoring workspace
Source baseline	v3.0 requirements baseline (8 Oct 2026); all prior versions (v1.0–v2.5) folded in and reconciled

Version History

Version	Date	Change
4.0	**14 Oct 2026**	**Clean-slate rewrite.** All v2.x carried-forward requirements rewritten in unified v3.0/multi-domain vocabulary. Redundancies removed: PLC-006 retired (subsumed by PLC-013); PLC-004/ PLC-009 amended to reference domain groups and PLC-015 pipeline; PLC-003 seed-record fields updated with CRV/DPB/domain token; PLC-007/008 explicitly state customer-aggregate scope; PLC-011 amended with per-domain scope note (PLC-016); DR-002/003/004/018 re-scoped with domain qualifier; CAP-023 reconciled with CAP-025; FIN-001/004 include domain dimension; A.1–A.9 reconciled with A.12 in unified Formulas & Calculation Reference section. No requirement content added or removed — only harmonised. Hard Constraints unchanged (HC-1..HC-11).
3.0	8 Oct 2026	Multi-domain customer model folded in. Domain-group CRGs per environment & region; two-scope region selection (CRV customer-scoped, DPB domain-scoped); DR Foundation; forward-items evaluation. Added PLC-013..016, DOM-001..004, CAP-025, DR-021, QUA-015, C-14/C-15/ C-16. Carried-forward requirements verbatim from v2.5; only Section 11 and dependent sections amended.
2.5	24 Sep 2026	Foundation aligned to Microsoft Learn. Sharing stays

Version	Date	Change
		Preview; QUA-013 documented; two-cap quota model; CAP-023 zonal CRG structure; CAP-016 logical zone numbers per subscription.
2.5 (amended)	7 Oct 2026	γ fairness denominator defined — PLC-012 geography customer count.
2.4 (amended)	11 Sep 2026	Middle East DR offering resolved — cross-geography DR to Europe. DEC-001 decided. Added DR-020, PLC-010b.
2.4	7 Sep 2026	Reservation-model gaps folded in. Added CAP-020..024, PLC-011, OPS-006, C-12/C-13.
2.3	7 Sep 2026	Region scope expanded to five geographies. PLC-010a generalised to all two-region geographies.
2.2	27 Aug 2026	EU cross-geo DR region corrected to Switzerland North.
2.1	27 Aug 2026	Reciprocal multi-source DR hosting, non-concurrent max-sizing, DR index, standby activation, CVAL/DR co-location.
2.0	27 Aug 2026	Pivot to minimal bootstrap + dynamic reconciliation; separated capacity vs quota; distributed DR; seed-record model.
1.0	22–26 Aug 2026	Initial capacity reservation design flow.

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1. Purpose & Business Context

This document defines the functional, operational, governance, security, observability, cost, and non-functional requirements for managing **Azure VM capacity reservations** and **regional VM quota** across the platform estate.

The target solution is a **Capacity & Quota Management Engine** that:

- protects production deployment capacity as the first priority;
- provides a **configurable bootstrap** position for disaster recovery instead of a fixed large reserve;
- **dynamically reconciles** reserved capacity against actual running (allocated) demand;
- **pools and allocates** regional quota across hundreds of subscriptions;
- supports **distributed DR** across multiple regions in a geography;
- supplies **fresh capacity state** to the region-selection and AEP provisioning workflows;
- minimises idle reservation cost without weakening required production and DR controls;
- treats each customer as **one placement identity with many capacity lanes** — region selection is bound once per customer (Customer Region Vector, CRV), while Instance Groups and capacity are domain-group-scoped (Domain Placement Binding, DPB); and
- produces auditable evidence for operations and compliance (e.g., SOC 2 DR assertions).

Why this document exists. The prior write-up was a design flow, not a requirements document. Because business direction changes frequently — regions, DR scope, and customer commitments shift regularly — the team agreed to maintain a **working requirements document** as the authoritative reference that the capacity engine is built against, evolving it as decisions firm up.

v4.0 — clean-slate rationale. v3.0 folded the multi-domain customer model into the baseline but carried v2.x requirements verbatim, creating redundancies (e.g., PLC-006 subsumed by PLC-013, formulas A.1–A.9 restated at different granularity in A.12, DR-002..004 written at customer scope rather than domain scope). v4.0 rewrites every requirement in a single, unified voice — no legacy amendment layers, no version-tagged caveats — while preserving every requirement ID for traceability.

2. Strategic Drivers & Constraints

These drivers shape every requirement below.

- **Cost is a primary constraint.** Leadership is in a cost-reduction cycle; improving margins (and, ultimately, EBITDA ahead of an IPO) is an explicit organisational goal. Large idle DR reservations (estimated in the **millions of dollars per year**)

will not receive business approval, so DR must start lean and scale on demand.

- **Quota as a governor.** Per Melvin's strategy, **quota is used as a cost and consumption governor** — it caps how much capacity a team can consume. Production maintains a quota buffer to support growth; DR quota strategy must align to the capacity-sharing model.
- **Business volatility.** Requirements evolve continually (region scope has both narrowed and re-expanded over time — e.g., Japan East is currently pending confirmation, while Australia and Asia Pacific are now back in scope; DHL-dedicated and Apple lost). The design must be **flexible and configuration-driven** rather than optimised for a single fixed scenario — the in-scope region catalogue (Section 6) is a configurable item (REG-001).
- **Legal ownership of Middle East — DR now offered cross-geo (DEC-001 resolved).** Legal has taken charge of the Middle East programme. Following legal/business direction, **DR is now offered for Middle East via a cross-geography model**: Prod and CVAL are provisioned in a selected **Middle East** Standard region and **DR is placed in a selected Europe Standard region**, both chosen by the weighted capacity placement model (DR-020, PLC-010b). Data-sovereignty constraints remain material — the per-country/region `DR_NOT_OFFERED` flag (DR-014) remains available where legal still prevents an acceptable DR design.
- **Region strategy (current).** Five geographies are in scope — **US, Europe, Australia, Asia Pacific, and Middle East** (Section 6). US runs a three-region distribution model; Europe, Australia, and Asia Pacific run a two-region (CVAL/DR co-located) model; **Middle East runs a cross-geography DR model** (Prod+CVAL local in Middle East, DR in Europe — DR-020, PLC-010b). The catalogue is configuration-driven and expected to keep changing.
- **Multi-domain customer model.** A customer may own multiple **domain groups** (e.g., TMS, WMS). Region selection is customer-scoped (one CRV per customer × geography, reused by all domains). Capacity and instance groups are domain-scoped (one DPB per domain × environment × region). This two-scope separation is the structural core of the engine's placement pipeline.
- **No multi-cloud.** Multi-cloud DR has been repeatedly rejected at the ELT level; "all of Azure down" is explicitly out of scope.
- **Preview-feature dependency. Capacity Reservation Sharing** is a preview feature. Production placement does not depend on it. Until sharing is generally available for the API version in use, a production VM is placed on a reservation in the **deploying subscription** (CAP-013, DEP-001).

3. Design Principles

1. **Production protection first.** Prioritise production availability and the next approved production deployment above all else.

2. **Quota and capacity are separate resources.** Manage them independently, but validate their relationship before any deployment.
3. **Allocated demand drives reservations.** Reservation targets are computed from **allocated (running)** VMs plus a buffer — never from associated-but-deallocated VMs alone.
4. **Buffers are configurable.** Production and DR buffer values are policy inputs, not hard-coded constants.
5. **DR starts lean.** DR holds only the approved **bootstrap** capacity (enough to stand up control planes and begin recovery), not a fixed percentage copy of production.
6. **CVAL is a DR capacity source.** On a declared disaster, CVAL capacity may be shut down, disassociated, or reassigned per the approved runbook.
7. **Regional & zonal isolation.** Reservations and quota are region- and zone-bound; capacity in an unavailable region is not assumed reusable elsewhere.
8. **Placement uses current state.** Region selection uses a recent, authoritative capacity/quota snapshot; stale state must not drive placement.
9. **Cost is a first-class constraint.** Idle reservation spend is measured, attributable, reviewed, and tunable.
10. **Automation with guardrails.** Automated changes are scoped by approved configuration, validated before execution, logged, and reversible where Azure permits.
11. **Customer placement is seeded once.** The first approved production-region decision becomes the authoritative seed for all later domain groups and environments for that customer.
12. **Design for changing policy.** Geography scope, DR sizing, buffers, SKUs, regions, domain taxonomy, and environment rules are all configuration-driven.
13. **Two-scope separation.** Region selection is customer-scoped (CRV); capacity binding is domain-scoped (DPB). A domain never re-runs region selection; a customer-level region change occurs only via the approved migration/exception workflow.

4. Terminology

Term	Meaning in this document
Allocated VM	A running VM currently consuming compute capacity.
Associated VM	A VM linked to a Capacity Reservation Group, whether running or deallocated.
Availability Set	Legacy Azure fault-/update-domain VM grouping. **Mutually exclusive with Capacity Reservations** — Availability-Set VMs are ineligible for reservation management (CAP-020) and must be redeployed into an availability zone before onboarding (CAP-021).
Available reserved capacity	Reserved capacity not currently consumed by allocated VMs.
Buffer target	Approved capacity held above current allocated demand for a defined scope.
Capacity Reservation Group (CRG)	Azure construct holding reservations for one region. A group is either **zonal** (zones declared at creation, immutable; one reservation per VM size per zone) or **non-zonal** (Azure pins the group to one zone at first reservation; VMs deploy without a zone). The engine organises CRGs per domain × environment × region (CAP-025).
Core subscription	A shared platform/”core” subscription whose VMs are **all classified production** for reservation and buffer purposes (CAP-001a); non-production workloads must not run there (ENV-003).
Customer Region Vector (CRV)	Customer-scoped, cached region binding `(geography, distribution model, prod, cval, dr)` decided once per customer × geography and reused by **all** domain groups (PLC-013).
CVAL	Customer validation environment; its capacity may contribute to DR readiness.
Domain group	A workload-domain grouping of a customer’s capacity (`wl-*` token, e.g. TMS, WMS) sharing one customer seed row but owning **separate instance groups** (ig) per environment/region (PLC-014, DOM-001/002). Domain is

Term	Meaning in this document
	the unit of ig/capacity binding; the customer is the unit of region binding (PLC-013).
Domain Placement Binding (DPB)	The deterministic per-domain capacity binding `DPB(D,E)` that resolves ig subscription, CRG set, CR sizing, zone distribution, and quota draw for one domain × environment × region, using the pre-bound CRV region (PLC-014).
DR bootstrap capacity	Minimum deployed/reserved platform capacity required to initiate recovery orchestration.
DR Foundation CRG set	The dedicated DR-bootstrap CRG set of a domain in its DR region — structural base capacity, always present with pre-staged quota; extendable on failover via the domain's own CVAL/DR shared CRG (CAP-025, DR-021).
Instance group (ig)	The subscription-bound capacity set that materialises **one domain × environment × region**: its subscription(s), CRG set (CAP-025), and quota draw. ig is domain-group-scoped; the region of an ig always comes from the customer's CRV (PLC-014).
Seed reservation	A managed reservation created at reserved quantity **0** for an eligible SKU × region × availability zone, as part of the initial **seed matrix** (CAP-022); reconciliation scales it up from 0 when allocated demand first appears. Distinct from the placement seed record below.
Seed matrix	The full set of count-0 **seed reservations** across every eligible SKU/AZ combination in the scope file (CAP-022), governed by product-team budget approval.
Consumer subscription	Subscription where a VM deploys and consumes a shared reservation owned elsewhere.
Provider subscription	Subscription that owns a reservation shared with other subscriptions.
Quota pool / quota group	

Term	Meaning in this document
	An Azure Quota Group moves unused quota onto a member subscription. Deploy-time checks that subscription's **VM-family** vCPU and **total regional** vCPU, not the group limit. One subscription belongs to one group; each transfer is scoped to one region and one VM family (QUA-003, QUA-004). All of a customer's domain ig subscriptions belong to a centrally managed quota group per region × VM-family cell (QUA-015).
Quota hoarding	Governance practice of collecting default per-region quota into a family pool for controlled reallocation.
Seed record (a.k.a. **placement seed**)	Authoritative customer record holding: customer/realm identifier, geography, distribution model, production region, CVAL region, DR region (or `NOT_OFFERED`), CRV vector (PLC-013), domain groups and their DPB references (PLC-014), products covered, decision timestamp, policy/engine version, capacity-snapshot reference, exception reference (if any), and approval metadata. Distinct from a seed reservation — "seed record" is about where a customer is placed; "seed reservation" is a zero-count reservation object waiting to scale.
Total customers (geography)	Count of distinct customers with at least one environment (Prod, CVAL or DR) provisioned or being deployed in any region of a geography. Denominator of the γ distribution-fairness term; one live count per geography, shared by all environments (PLC-012, F.10).
SKU scope	The SKU/VM-family × region × zone × subscription × environment combination managed by policy.
AEP	The provisioning/automation entry point that triggers region selection and deployment. AEP deployments carry a domain group token (DOM-001) in addition to customer/realm, product, environment, and region inputs (INT-003/INT-008).

Term	Meaning in this document
Source region	A production region whose workload fails over on outage.
Destination (DR) region	A surviving region hosting standby DR instances for one or more source regions.
Non-concurrent sources	Multiple source regions that (per DR-001) are assumed never to fail simultaneously.

5. Scope

5.1 In Scope

- Azure VM Capacity Reservations and Capacity Reservation Groups.
- Regional and VM-family quota inventory, pooling, allocation, reclamation, and monitoring.
- Production, CVAL, and DR compute-capacity policies.
- Cross-subscription capacity reservation **sharing** where supported and approved.
- Capacity/quota state consumed by region selection and AEP provisioning.
- Multi-domain customer model: customer-scoped region selection (CRV) and domain-scoped capacity binding (DPB).
- Automated reservation reconciliation.
- Cost, compliance, and operational evidence.
- Alerting, dashboards, audit records, and exception workflows.
- Capacity rebalancing for new deployments, growth, shutdowns, and DR exercises.

5.2 Out of Scope (this baseline)

- Application-level data replication / failover implementation.
- Database and non-VM PaaS capacity mechanisms.
- Multi-cloud disaster recovery.
- Full AEP redesign beyond the required integration contracts.
- Per-country legal exceptions within Middle East that may still warrant `DR_NOT_OFFERED` (DR-014) despite the geography-level cross-geo DR offering now in scope.
- Permanent R&D reservations (short-lived POC reservations may be supported).
- “Entire Azure region-set / global Azure outage” as an addressable failure mode.

6. Region Strategy & Classification

The estate is organised into two capacity classes across the in-scope regions. Five geographies are in scope: **US, Europe, Australia, Asia**

Pacific, and Middle East. **US** operates a **three-region distribution model**; **Europe, Australia, and Asia Pacific** operate a **two-region distribution model** (see REG-003 and the ENV-003 CVAL/DR co-location mechanism). **Middle East** uses a **cross-geography DR model** (DR-020, PLC-010b) — Prod and CVAL in a selected Middle East Standard region and **DR in a selected Europe Standard region**.

Class	Behaviour	Notes
Standard capacity regions	Eligible for automatic selection by the placement/region-selection engine.	Default provisioning targets for production, CVAL, and DR.
Restricted deployment regions	Production-only; **not** auto-selected unless explicitly provided as the production region.	Used only when a customer/contract specifies the exact region.

In-Scope Region Catalogue (configurable — see REG-001)

Geography	Distribution model	Standard capacity regions (engine-selectable)	Restricted regions (exception only)	Notes
US	3-region	West US 3 · Central US · Canada Central	East US 2 (exception required)	Only geography with full Prod / CVAL / DR region separation.
Europe	2-region	Switzerland North · Sweden Central	North Europe · West Europe (exception)	Also serves as the **cross-geo DR destination geography for Middle East** (DR-020); the specific Europe DR region is weighted-selected, not fixed (REG-002).
Australia	2-region	Australia East · Australia Southeast	—	In scope as of v2.2.
Asia Pacific	2-region	East Asia · Southeast Asia	—	**Japan East** — pending business confirmation before inclusion as a Standard region.
Middle East	Cross-geo DR	Saudi Arabia Central · UAE North (host Prod+CVAL, co-located)	—	**DR offered cross-geo to Europe** (DEC-001 resolved; DR-020, PLC-010b). Per-country `DR_NOT_OFFERED` (DR-014) retained for legal carve-outs.

All Standard capacity regions in every geography are eligible to host **Prod, CVAL/NonProd, and DR** environments; region selection is driven by the placement scoring pipeline (PLC-007/PLC-008 at **customer-aggregate** scope per F.11) over the regions currently available in that geography.

REG-001 — Configurable region catalogue. The region classification, eligibility, distribution model, and per-region flags (e.g., `DR_NOT_OFFERED`, zone support, restricted) must be configuration-driven and versioned. Adding, removing, or reclassifying a region is a configuration change, not a code or design change.

REG-002 — Example correction discipline / cross-geo DR region is weighted-selected. Region examples must be sourced from authoritative configuration, not slideware. The Middle East cross-geo DR destination is not fixed to any single region; the engine selects it via the weighted capacity placement model (DR-020).

REG-003 — Distribution model. Three distribution models are in scope, selected per geography by configuration (REG-001):

1. **Three-region model (US).** Prod, CVAL, and DR each occupy a distinct region — best distributes customer workloads and minimises the DR capacity reserved per region.
2. **Two-region co-located model (Europe, Australia, Asia Pacific).** A two-region geography cannot separate all three environments, so Prod occupies one region and **CVAL + DR co-locate in the other** (PLC-010/PLC-010a). This is a supported, normative configuration provided co-location capacity accounting (HC-6, HC-7, PLC-010) is honoured.
3. **Cross-geography DR model (Middle East).** Prod and CVAL co-locate in a selected **Middle East** Standard region (separate CRGs per ENV-003), and **DR is placed in a selected Europe Standard region** chosen by the weighted capacity placement model. CVAL co-locates with **Prod locally**, not with DR; the CVAL-sacrifice DR bootstrap (DR-005/DR-006) does **not** apply, and Middle East DR is served by **dedicated reserved DR capacity in Europe** (DR-020, PLC-010b).

Expanding a geography to three-plus regions remains a design goal.